

DGA Status Report

Dissolved gas analysis — status assessment, IEEE C57.104-2019 Figure 2
Biyagama IBT 02 (R) - certificate check

Status 3 - Probably suspicious

Report generated	2026-09-07 13:16	Latest sample	2024-12-01	Samples used	2	History from	2024-06-01
O2/N2 section	>0.2	Age band	>30 (43 yr)	Table 4 period	not applicable	Multi-point rate	not available

1. Verdict

Status 3 on H2 above the Table 2 (95th percentile) level.

Required action — Identify the fault (Duval triangle and pentagon, Rogers ratios), place the unit under increased surveillance, add supporting tests and consult a transformer specialist.

2. What drove the status

- H2 is 125 ppm, above the Table 2 (95th percentile) level of 90 ppm.
- H2 rose +38 ppm since 2024-06-01, against a Table 3 allowance of 25.
- C2H2 rose +1 ppm since 2024-06-01, against a Table 3 allowance of any increase.

3. Decision path (Figure 2)

Step	Question	Answer	Detail
Levels vs Table 1	Is every measured gas below its 90th percentile level?	No	H2 above
Change vs Table 3	Is every change since the previous sample within the allowance?	No	H2, C2H2 over
Rate vs Table 4	Is every multi-point rate within the allowance?	Not applicable	no multi-point rate available
Levels vs Table 2	Is any measured gas above its 95th percentile level?	Yes	H2 above
Result	Figure 2 outcome	Status 3	Probably suspicious

4. Gas-by-gas evidence

Gas	Latest ppm	T1	T2	Level	Change since previous	T3	Verdict	Rate ppm/yr	T4	Rate
H2	125.0	40	90	above T2	+38.0	25	over	—	—	n/a
CH4	5.0	20	30 ‡	within	+5.0	10	within	—	—	n/a
C2H6	0.0	15	40 ‡	within	0.0	7	within	—	—	n/a
C2H4	3.0	60 ‡	125 ‡	within	+1.0	n/a ‡	n/a	—	—	n/a
C2H2	2.0	2	7	within	+1.0	any incr.	over	—	—	n/a

Not measured, so excluded from every comparison: CO, CO2.

5. Limits applied

Column selected by the standard: O2/N2 >0.2, age >30, Table 4 not applied.

Table	H2	CH4	C2H6	C2H4	C2H2	CO	CO2
Table 1 — 90th percentile level	40	20	15	60	2	500	5500
Table 2 — 95th percentile level	90	30	40	125	7	600	8000
Table 3 — change between samples	25	10	7	n/a	any incr.	175	1750
Table 4 — rate (ppm/yr)	not applied	not applied	not applied	not applied	not applied	not applied	not applied

6. Sample history used

Date	H2	CH4	C2H6	C2H4	C2H2	CO	CO2	O2	N2
2024-06-01	87	0	0	2	1	—	—	—	—
2024-12-01	125	5	0	3	2	—	—	—	—